www.tntextbooks.in

or $\frac{P}{T} = \text{Constant } k$

If P_1 and P_2 are the pressures at temperatures T_1 and T_2, respectively, then from Gay Lussac's law

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

Activity -1

The table below contains the values of pressure measured at different temperatures for 1 mole of an ideal gas. Plot the values in a graph and verify the Gay Lussac's law. [Lines in the pressure vs temperature graph are known as iso chores (constant volume) of a gas.]

Temperature (in deg celcius)	32	69	94	130	154	191
Pressure in 50 L container (atm)	0.51	0.56	0.6	0.66	0.7	0.76
Pressure in 75 L container (atm)	0.34	0.37	0.40	0.44	0.47	0.51

6.2.4 Avogadro's Hypothesis

Avogadro hypothesised that equal volumes of all gases under the same conditions of temperature and pressure contain equal number of molecules. The mathematical form of Avogadro's hypothesis may be expressed as

$V \alpha n$,

$$\frac{V_1}{n_1} = \frac{V_2}{n_2} = \text{constant} \text{ -------- } (6.10)$$

where V_1 & n_1 are the volume and number of moles of a gas and V_2 & n_2 are a different set of values of volume and number of moles of the same gas at same temperature and pressure.

Evaluate Yourself

4. An athlete in a kinesiology research study has his lung volume of 7.05 dm^3 during a deep inhalation. At this volume the lungs contain 0.312 mole of air. During exhalation the volume of his lung decreases to 2.35 dm^3. How many moles of air does the athlete exhale during exhalation? (assume pressure and temperature remain constant)

6.3. Ideal gas equation

The gaseous state is described completely using the following four variables T, P, V and n and their relationships were governed by the gas laws studied so far.

Boyle's law $V \alpha \frac{1}{P}$

Charles law $V \alpha T$

Avogadro's law $V \alpha n$

We can combine these equations into the following general equation that describes the physical behaviour of all gases.

$$V \alpha \frac{nT}{P}$$

$$V = \frac{nRT}{P}$$

where, R is the proportionality